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Enhanced Steam Plan Optimization for a Thermally Developed Heavy Oil Field in Kuwait

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Abstract

To maximize the value of limited steam capacity and achieve short-term production targets for a thermally developed project in the north of Kuwait, an automated workflow was developed. This workflow optimizes both steam distribution and operating modes—Cold, Cyclic Steam Stimulation (CSS), or Steam Flooding (SF)—for 35 active patterns, encompassing over 1,000 wells.

The automated workflow begins by extracting historical injections and production data from the corporate database. It then generates three performance-based production forecasts (Cold, CSS, and SF) for each pattern using historical data and reservoir characteristics, which are validated against simulation results. An optimization tool evaluates the optimal combination of operating modes based on defined constraints, aiming to maximize production while adhering to steam capacity and facility constraints. The workflow produces an optimized steam plan and corresponding production forecast from multiple steaming scenarios within minutes, significantly accelerating and supporting the decision-making process.

The steam optimization tool successfully generated an optimal steam plan for a 100 MBCWEPD steam capacity across the 35 active patterns. Additionally, the tool enabled rapid evaluation of multiple development opportunities. Key findings from these assessments include prioritizing newly commissioned patterns, reducing steam rates in SF patterns to reallocate steam more effectively, and identifying additional pattern conversion opportunities, all contributing to cost reduction and production enhancement.

This study introduces a novel, efficient approach to steam optimization, enabling faster and more informed decisions. By optimizing steam allocation and distribution, the automated workflow helps reduce operational costs and maximize short-term production. The adopted methodology significantly improves planning through automation and can be effectively extended to other thermal recovery projects.

Introduction

The Lower Fars Formation of South Ratqa Field located in the north Kuwait, is home to the largest thermal development project currently operational in the Middle East. The formation holds heavy oil in a shallow, unconsolidated sandstone reservoir and consists of two main sequences, each subdivided into A and B units.

These sub-units are typically, though not always, separated by discontinuous low-permeability layers that restrict vertical fluid flow.

While the fluid properties, rock characteristics, and geological features of the Lower Fars Formation vary across the field, three distinct zones have been identified. The primary differentiating factors among these zones are the viscosity of the oil in the Upper-B unit and the presence of bottom water.

Due to the heavy and viscous nature of the oil, along with other reservoir's characteristics, the South Ratqa Field is particularly well-suited for thermal recovery methods. The South Ratqa Phase-1 development has been designed to produce 60 thousand barrels of oil per day (MBOPD), supported by an initial steam generation capacity of up to 100 thousand barrels of cold-water equivalent per day (MBCWEPD). The development employs Cyclic Steam Stimulation followed by steam flooding, using vertical wells arranged in inverted 9-spot configuration with 20-acre spacing¹. The field is divided into two main blocks, each subdivided into surface patterns, with each pattern containing at least nine inverted 9-spot sub-patterns.

In Phase-1, oil production is currently limited to the two uppermost oil-bearing reservoir layers. Production from deeper intervals, where present, has been deferred to future development phases. Although steam flooding has commenced in select patterns, full-field conversion to steam flooding is contingent on expanding injection capacity, which has experienced delays.

Maintaining production at target levels in the South Ratqa project is highly dependent on the volume of steam injected. In scenarios where steam supply is insufficient to meet the demands of all active patterns, the strategic allocation of available steam becomes critical. To address this, the reservoir management team has developed a workflow aimed at optimizing the use of limited steam resources. This approach is vital for sustaining reservoir pressure and ensuring optimal performance of the most economically valuable production patterns.

Steam Plan Optimization Objectives

Delays in steam injection have been the primary obstacle in maintaining targeted production levels at the South Ratqa Field. In the initial development phase, steam injection capacity for the first 24 CSS patterns were estimated at approximately 100 MBCWEPD. The original plan envisioned transitioning from CSS to SF after three CSS cycles (approximately three years), supported by expanded steam generation and water handling capacities. However, delays in implementing these infrastructure projects heightened the importance of steam injection optimization to sustain production.

In certain low-viscosity zones of the field, additional CSS cycles proved to be less effective, necessitating early conversion to steam flooding despite the limited steam capacity. The transition of eight CSS patterns to steam flooding raised subsurface steam demand. Yet, due to ongoing operational challenges, the actual average steam injection rates in 2022, 2023, and 2024 fell short of the planned 100 MBCWEPD. This shortfall adversely affected the performance of CSS patterns and contributed to an overall production decline across the South Ratqa area.

By promptly identifying and applying Well-Reservoir-Facility Management (WRFM) strategies—some of which align with industry best practices²—performance degradation was effectively mitigated. A notable example of this approach was the cessation of steam injection in underperforming patterns affected by bottom-water influence, allowing for the reallocation of steam to more productive areas. This optimization enhanced overall resource utilization efficiency.

In 2024, the commissioning of 11 new CSS patterns increased the total number of SR patterns to 35, raising the steam demand to over 150 MBCWEPD. As the gap between steam demand and available capacity widens, optimizing steam allocation has become critical to achieving production objectives. In response, the team developed an efficient, time-saving methodology for steam distribution across SR patterns. This approach also guides the selection of optimal operating modes—Cold, CSS, or SF—for each active pattern, with a focus on maximizing short-term production.

The CSS steam injection strategy is designed to address uneven pressure depletion, enhance short-term output, and ensure uniform heat distribution across reservoir patterns. When sufficient steam capacity is available to support all CSS and SF operations, the steam allocation process follows a streamlined three-step approach:

- Determine the volume of steam to inject into each pattern.
- Define the number of wells to be steamed in each CSS batch.
- Select the specific wells within each pattern for steam injection.

Under conditions of limited steam availability, a fourth step is introduced: prioritizing patterns for steam injection based on their expected production performance and return on investment. This involves assigning steam targets and determining the most effective operating mode (Cold, CSS, or SF) for each of the 35 patterns. Due to the operational variability and complexity of assessing production potential across all patterns, this prioritization process is both intricate and resource intensive. Traditionally, it has required multiple manual iterations and close collaboration between steam planning and production forecasting teams.

The patterns prioritization process begins with the extraction of historical injection and production data from the company database, followed by the generation of three production forecasts for each pattern—Cold, CSS, and SF—based on historical performance and the reservoir properties of each pattern. These forecasts are then validated against results from the most recent simulation studies. The optimization tool is subsequently utilized to evaluate the optimal combination of operating modes that maximize production while adhering to well and facility constraints within the available steam capacity. This approach enables the development of a steam injection plan along with an associated short-term production forecast for multiple scenarios, facilitating rapid steam optimization.

This workflow addresses the challenge of the substantial time typically required to generate forecasts for multiple scenarios using conventional simulation runs. It enables reservoir engineers to quickly identify the most effective steam plan to maximize short-term production. By integrating automated forecasting with steam planning, the tool significantly reduces the time required to optimize steam allocation and generate associated forecasts, supporting faster and more informed decision-making to achieve short- and mid-term production targets.

Table 1 presents the list of required input parameters used to generate pattern-level production forecasts under various operating modes, where the initial operating mode for each pattern is also defined.

Table 1—Definition of Input Parameters for Generation of Pattern Forecast under different Modes of Operations: (a) Cold; (b) CSS; (c) SF.

Pattern ---->	B-03/P-22	Assumptions		B-02/P-04	B-02/P-07	B-02/P-08	B-02/P-09	B-02/P-10	B-02/P-13
CD or PD:	CD	Cold	Exp decline Coefficient	0.60	0.60	0.60	0.60	0.60	0.60
Forecasting Starts:	3/1/2025		Cold WC	0.35	0.35	0.35	0.35	0.35	0.35
		CSS / CSF	SOR	1.90	2.20	1.80	1.80	1.80	2.20
CSS TFSR	1.2		Post Plateau Monthly Decline	0.975	0.975	0.975	0.975	0.975	0.975
			Ramp-up months	10	10	10	10	10	10
First 2 Cycles Prod Adj	2.2		Plateau + Ramp-up months	18	16	18	18	18	18
First 2 cycles Decline Adj	0.9		1st Cycle Date (if not started)						
		Steam Flood	1st Cycle Prod Duration						
			SOR @ 7500 bpd Steam	2.4	2.6	2.3	2.2	2.4	2.8
SF TFSR	1.1		Monthly Decline	0.99	0.99	0.99	0.99	0.99	0.99
			Ramp-up months	10	10	10	8	8	10
			Plateau + Ramp-up months	24	24	24	24	24	24
			last 1.5-year CSOR (for ref)	2.33	0.86	2.19	2.18	2.19	1.04
		Picked mode		3	2	3	3	3	3

Patterns Production Forecasts

South Ratqa production forecast is heavily influenced by steam injection rates and the allocation of steam across the 35 active patterns. To prioritize patterns awaiting steam, it is essential to understand how each pattern responds to various steam injection targets and operating modes. The steam optimization process involves generating three distinct production forecast profiles for each pattern, corresponding to the different operating modes: Cold, CSS, and SF. These profiles are derived from historical data and reservoir characteristics, employing performance-based methods with options for adjustment. In some instances, production profiles from simulation runs may also be incorporated, though relying on simulation models can be time-consuming and often require multiple iterations.

By combining production profiles for all active SR patterns, various scenarios can be created. The tool then selects the optimal combination of operating modes for each pattern, necessitating the use of advanced optimization features to efficiently determine the best steam allocation and operational configuration.

Reservoir Properties Consideration. The reservoir quality in the Lower Fars layers exhibits significant variation across the different patterns within the South Ratqa Phase-1 Area. Figure 1 highlights the main fluid and reservoir properties, showing how these characteristics vary across different patterns. Oil production in certain patterns is particularly challenging due to the presence of a bottom water layer. The Bottom Water zone is absent in the northern half of the SR Phase-1 Area but gradually increases to the south, reaching thicknesses of 25 feet or more. Additionally, the SR Phase-1 Area contains intervals of Top Gas within the reservoir zones. These top gas pools are predominantly concentrated in the western half of the SR Phase-1 Area (not shown in Figure 1).

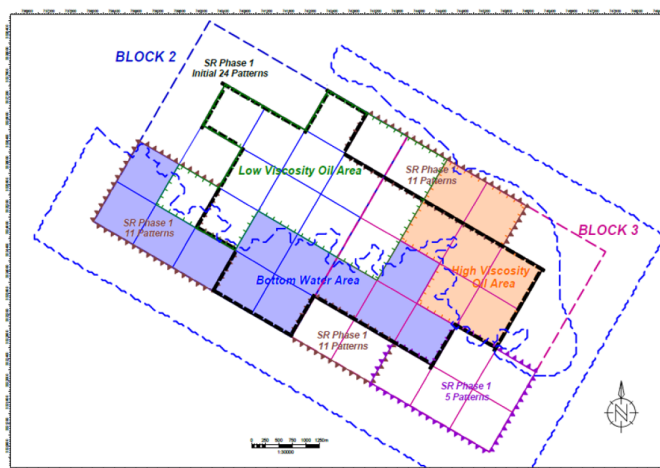


Figure 1—Location of Patterns with Distinct Geological Features

Beyond these formation parameters, the complexity of production in the SR Phase-1 Area is further compounded by highly variable reservoir fluid properties. Oil viscosities generally increase from the northwest to the southeast, with the viscosity gradient in the lower-producing layers being notably steeper. In the southern and eastern regions of the SR Phase-1 Area, poorer steam injectivity and lower oil production were observed during the initial steam cycle. In addition to variations in oil viscosity, oil densities also show regional differences across the SR Phase-1 Area, typically increasing from the northwest to the southeast.

The variation in reservoir properties between patterns must be considered when determining the forecasting parameters outlined in the following sections. These variations—such as differences in oil viscosity, density, and the presence of bottom water or top gas—significantly influence the behavior of each pattern and are essential for developing accurate production forecasts. As such, the forecasting parameters are tailored to reflect the unique characteristics of each pattern, ensuring that production predictions and steam allocation are as precise and optimized as possible.

Cold Production Forecast. The Decline Curve Analysis (DCA) technique is employed to predict declining production rates based on historical production data from patterns that were producing without steam. This method enables the estimation of future production performance by analyzing the rate at which production declines over time.

To account for the increase in oil viscosity as heat dissipates into the formation and the reduction in pressure as production progresses, an exponential decline model was applied to all patterns, as outlined below:

$$q = q_i e^{-d(t-t_i)}$$

Where:

- q is the predicted oil rate at time t .
- t_i is the start of cold production decline (i.e. forecasting date).
- q_i is the initial production rate (used the average of the last three months production rate prior to forecasting date)
- d is the exponential decline coefficient.

The primary parameter to adjust for the cold production forecast is the exponential decline coefficient. This coefficient dictates the rate at which production declines over time due to factors such as increasing oil viscosity, reducing pressure, and other reservoir conditions in the absence of steam injection. By modifying this coefficient, the forecast can more accurately reflect the observed decline in production for each specific pattern.

In practice, the exponential decline coefficient is adjusted to align with historical decline trends observed in the pattern's production data. Additionally, simulation results can be used to further refine the coefficient, ensuring it accurately accounts for potential interferences between patterns. By utilizing simulation production profiles, the coefficient can be calibrated to incorporate these interferences, thereby enhancing the accuracy of the cold production forecast.

The gross production forecast is generated by taking the last known gross production value and applying the oil production decline model to project future values. If no gross production data is available for the most recent month, an initial water cut is assumed for each pattern. This assumption can be adjusted as needed in the Cold parameters' section (Table 1). The water production forecast is then calculated as the difference between gross production and oil production.

An example of the resulting cold production forecast for one of the 35 patterns is shown in Figure 2. This figure illustrates how both oil and liquid production decline under no steam injection conditions.

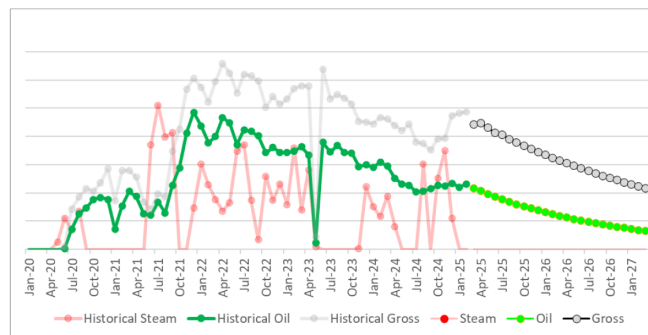


Figure 2—Sample Production Profile Generated for a Pattern under Cold Mode (No Further Steam Injection)

CSS Production Forecast. The initial results from the first CSS cycle review confirmed that the performance of patterns could be categorized into three main groups. The key differentiators in pattern performance are oil viscosity and the presence of bottom water, both of which significantly influence steam injection efficiency and oil recovery rates.

Parameters for CSS production forecast generation, are outlined in Table 1 under the CSS parameters' section. After selecting a pattern and/or modifying a parameter, the injection production plot for CSS is updated in real-time, as shown in Figure 3. This dynamic visualization enables users to quickly assess the impact of any changes on the forecasted production.

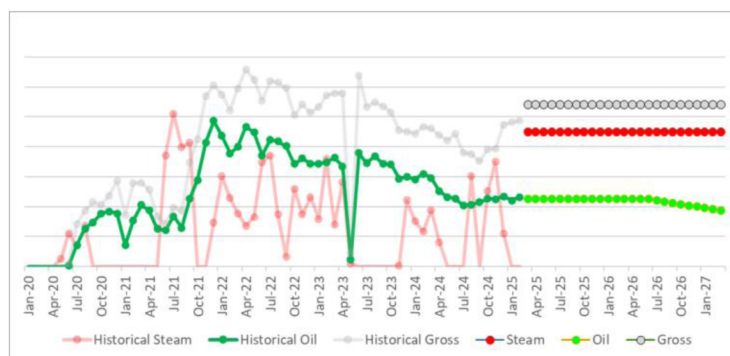


Figure 3—Sample Production Profile Generated for a Pattern under CSS Mode

For each pattern under CSS, a target steam injection rate is assumed based on historical reservoir performance, pressure, and geological characteristics. The assumed CSS steam injection target is combined with the expected Steam to Oil Ratio (SOR) to generate the oil production profile. Continuous CSS operation per is assumed per pattern (3-5 wells per pattern at a time), reflecting the ongoing field operations. The SOR, derived from historical CSS performance, provides a realistic estimate of the steam required to produce a unit of oil. If the most recent historical oil production rate significantly deviates—either higher or lower—from the calculated oil rate (derived from the steam injection target and SOR), a gradual production ramp-up or decline is automatically applied.

The CSS production plateau is anticipated to occur after an initial ramp-up phase, followed by a decline in production. The plateau duration is an adjustable parameter that accounts for historical production; if the plateau has already started, the forecast will show an earlier decline in production. Additionally, the post-plateau production decline rate can be adjusted based on historical performance and the latest simulation results, ensuring that the decline is accurately represented according to each pattern's specific behavior.

To account for the particularities of the first two CSS cycles in new patterns, adjustment factors have been applied to production, cycle duration, and production decline rate. These adjustments are clearly highlighted in the CSS parameters table, for proper consideration of the early stages of CSS operation.

Finally, total liquid production from CSS patterns is calculated using the steam injection profile and an assumed Total Fluid to Steam Ratio (TFSR). Water production is determined by subtracting the oil production from the total liquid production, providing a comprehensive view of the expected output for each pattern under CSS.

Steam Flood Production Forecast. The Steam Flood (SF) production forecast is automatically generated using the SF input parameters. Similar to the CSS forecast, the SF steam injection target is combined with the Steam to Oil Ratio (SOR) to generate the oil production profile. The expected SOR is determined based on historical pattern performance, reservoir properties, and the latest simulation results. If the most recent historical oil production rate deviates significantly from the rate derived from the steam injection target and SOR, a gradual production ramp-up or decline is applied to the forecast, ensuring a smooth transition.

Upon selecting a pattern and/or adjusting a parameter, the injection/production plot for SF is updated. This feature enables users to visualize changes instantly and assess the impact of different parameters on the forecast. Figure 4 presents a forecast generated for a pattern.

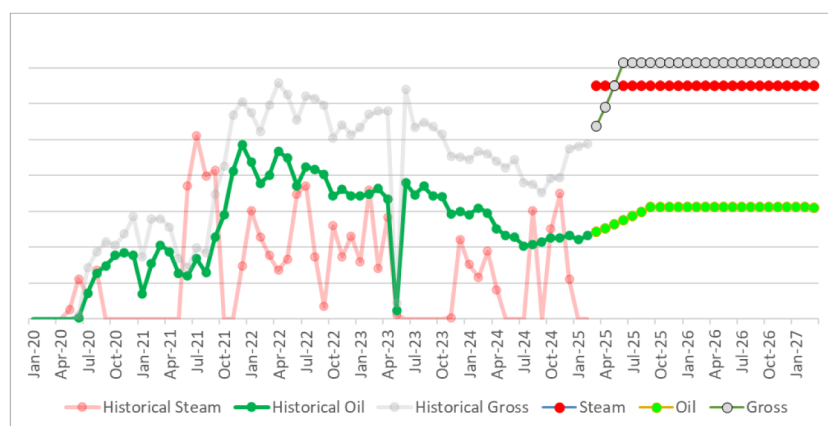


Figure 4—Sample Production Profile Generated for a Pattern under SF Mode

It is important to note that the SOR is expected to vary with different steam injection rates. To account for this, the SOR user input in the parameters table is automatically adjusted to reflect the assumed SF injection rate. The variation in SOR with respect to steam injection rate (Figure 5) was derived from multiple simulation runs and a trial conducted on two patterns, where steam injection was reduced by ~ 20%, and several production tests were performed to capture the SOR behavior at varying injection rates.

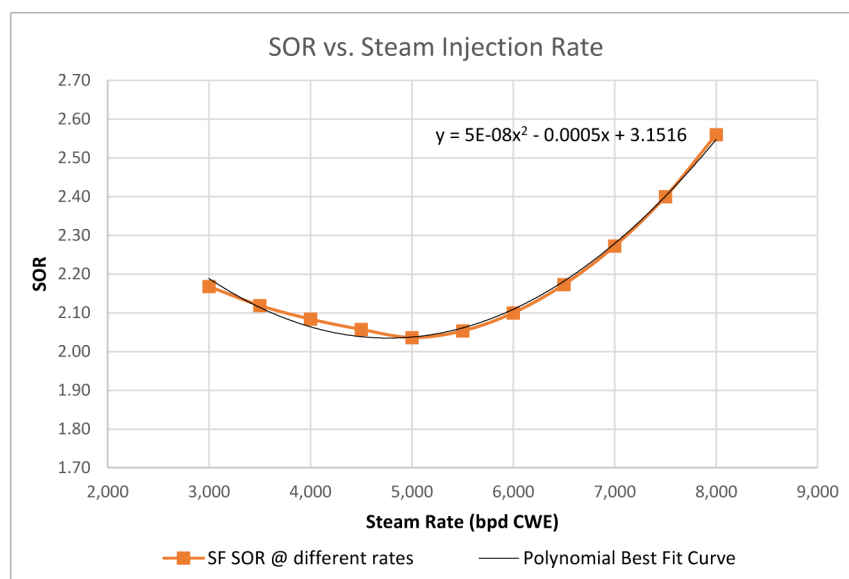


Figure 5—Expected Impact of Steam Injection Rate on the Steam-to-Oil Ratio for Patterns under Steam Flood Operations

Both the duration of the production plateau and the post-plateau production decline rate are adjustable in the SF parameters table to account for the expected SF production decline. These parameters are based on the historical performance of the pattern, supported by the latest simulation data. The production plateau duration incorporates the historical production profile; for example, if a plateau phase has already begun in the past, the forecast will account for an earlier onset of production decline after the forecast start date.

Total liquid production from SF patterns is calculated using the steam injection profile and an assumed Total Fluid to Steam Ratio (TFSR), which can be adjusted in the parameters table for each pattern.

Water production is derived by subtracting oil production from the total liquid production, resulting in a comprehensive production forecast that accounts for both oil and water output.

Aggregate Injection and Production Forecast. The operating modes are represented by numerical values ranging from 0 to 3, as seen in Table 2. This method of categorizing operating modes using numerical values enables efficient aggregation of production forecasts across the entire field. It supports optimized steam distribution and prioritization of patterns under varying operational conditions. By applying this approach, the tool can rapidly assess the impact of changes in operating modes and steam injection strategies on overall production performance.

Table 2—Definition of Constraints and Initial Operational Models for Various Patterns

Patterns -->	B-02/P-04	B-02/P-07	B-02/P-08	B-02/P-09	B-02/P-10	B-02/P-13	B-02/P-14	B-02/P-15	B-02/P-18	B-02/P-19	B-02/P-20
Optimize											
Current Operating Mode -->	3	1	3	3	3	3	3	3	1	3	1
Initial Operating Mode -->	3	2	3	3	3	3	3	3	1	3	1
Optimum Operating Mode -->	3	2	3	3	3	3	3	3	1	3	1
Legend	SF	CSS	SF	SF	SF	SF	SF	SF	Cold	SF	Cold
Forecasting Starts:	3/1/2025										
Assumptions / Constraints	0: OFF, 1: Cold, 2: CSS, 3: SF										
Min Operating Mode -->	3	1	3	3	3	3	3	3	1	3	1
Max Operating Mode -->	3	2	3	3	3	3	3	3	2	3	2

Once the operating mode for each pattern is selected, the forecasted injection and corresponding production rates—derived from the relevant forecasting data for Cold, CSS, or SF modes—are aggregated at the field level (Figure 6).

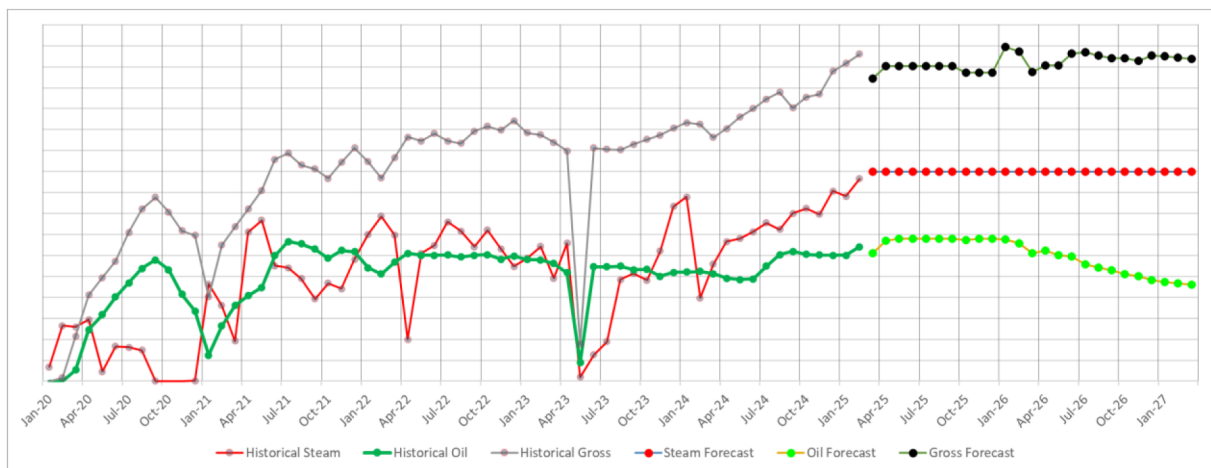


Figure 6—Example of the Resulting Field Injection and Production Profile After the Optimization Process

Steam Distribution and Operating Mode Optimization

The steam volume allocated to older CSS patterns plays a critical role in providing timely pressure support for heavily depleted patterns. Two steaming options are available for these patterns: Steam Flooding (SF), which requires a high steam injection rate per pattern, and Cyclic Steam Stimulation (CSS), which operates at a lower steam rate. However, under conditions of limited steam availability, deploying SF may reduce the steam available for other patterns, potentially leading to a decline in overall field production.

To maximize the value of constrained steam resources and achieve optimal short-term production, a robust prioritization process is essential. This involves determining which patterns should receive steam and selecting the appropriate operating mode (Cold, CSS, or SF) for each. To support this decision-making process, the Excel Solver optimization tool was integrated into the steam optimization workflow. The use of Solver allows users to efficiently prioritize patterns for steaming, ensuring that key business objectives—such as maximizing production or improving energy efficiency—are met, all while operating within well and facility constraints.

By leveraging Solver, the tool can identify the optimal combination of patterns and operating modes, dynamically adapting to changes in steam capacity, production targets, and operational conditions.

Facility Constraints. The oil production capacity of the South Ratqa Phase-1 facility is 60 MBOPD, with an additional 10% margin for operational flexibility—bringing the total capacity to 66 MBOPD. The oil processing facility is divided into two separate processing trains, each dedicated to a specific block: Block-2 and Block-3.

The steam distribution network originates from the Central Processing Facility (CPF), where steam is generated using Once-Through Steam Generators (OTSG). From there, two 24-inch and two 16-inch trunk lines transport steam to the respective blocks. Steam is then further distributed from these trunk lines to the individual patterns within each block.

For effective operations and maintain steam quality, the minimum steam flow assurance requirements are 22 MBCWEPD for Block-2 and 18 MBCWEPD for Block-3. These minimum flow rates are essential for preserving steam quality and ensuring the integrity of the flow lines. Figure 7 shows the facility layout, highlighting the injection and production streams.

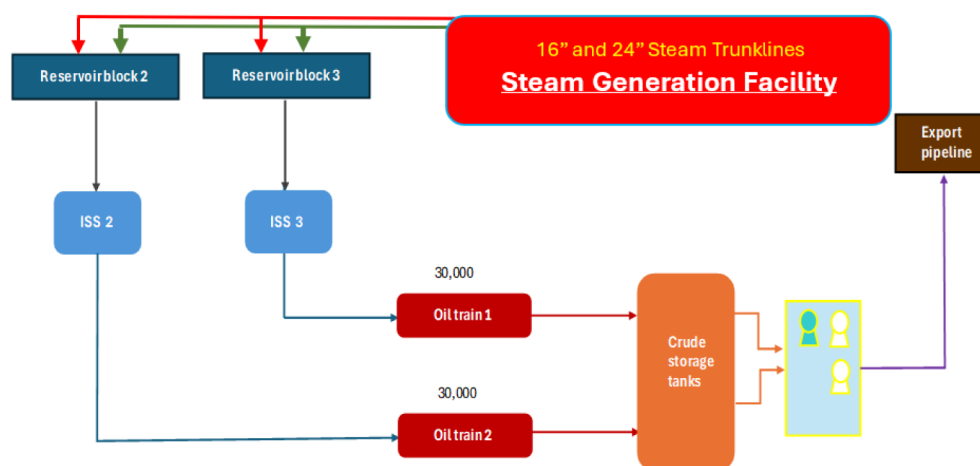


Figure 7—South Ratqa Facility Layout, Highlighting the Injection and Production Streams

Optimization Setup. The Microsoft Excel Solver add-in is used to determine the optimal (maximum or minimum) value of an objective function while satisfying a set of predefined constraints. This is done by adjusting a set of parameters known as decision variables, which influence both the objective function and the constraints. By modifying these variables, the solution that meets all constraints and achieves the optimal outcome is identified. In the context of steam optimization, Solver helps determine the best allocation of steam and selection of operating modes for each pattern, balancing production objectives with steam injection limitations.

For the steam optimization workflow, the objective function is defined as the average oil production rate over a user-specified period. This function is calculated during the optimization process to maximize short-term production, while adhering to two key constraints:

- **Minimum steam injection per SR block** – This constraint ensures that each steam injection block (Block-2 and Block-3) receives the required minimum steam volume to maintain flow assurance and steam quality.
- **Maximum steam capacity** – This constraint limits the total steam available for injection during the optimization period, based on the current operational capacity.

The decision variables in this optimization are the cells representing the operating mode of each pattern. These variables are constrained to integer values between 0 and 3, corresponding to the following operating states:

- **0: OFF** – No steam injection (pattern is shut-in).
- **1: Cold** – Pattern operates without steam (cold production mode).
- **2: CSS** – Pattern operates under Cyclic Steam Stimulation (CSS) mode.
- **3: SF** – Pattern operates under Steam Flooding (SF) mode.

Figure 8 illustrates the setup for the optimization process. The tool adjusts the operating mode (③ in Figure 8) of each pattern within the defined constraints (② in Figure 8) to maximize overall oil production (① in Figure 8). This approach ensures that both the total steam capacity and the minimum steam injection requirements for each block are satisfied during the optimization period.

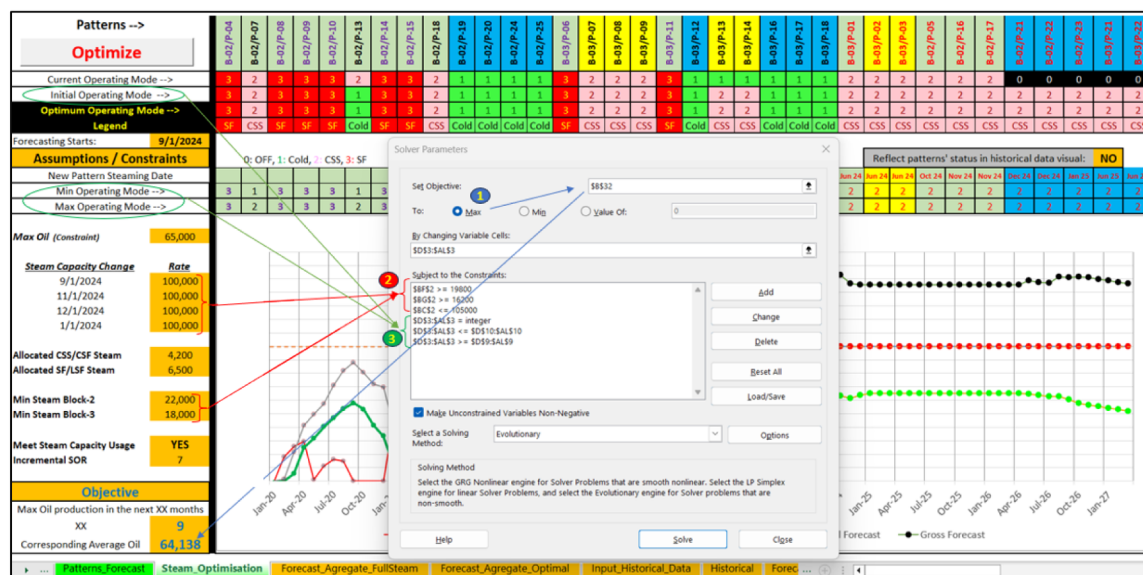


Figure 8—Optimization Setup with Objective Function and Constraints

Case Studies

Although additional steam is required to activate newly commissioned CSS patterns and restimulate southern patterns impacted by bottom water, the field continues to operate within its original steam capacity. This capacity was initially designed to support only 24 patterns under CSS operation. As fluctuations between steam demand and available capacity become more frequent, the optimization tool has proven essential in enabling faster adjustments to the steam plan, helping to sustain the South Ratqa production plateau.

While the optimization tool significantly improves steam injection planning, it should not be used in isolation or without due diligence. Some operational decisions—like restricting the conversion of a

pattern from steam flood (SF) back to CSS—require user input. The tool provides options to enforce these constraints, ensuring the optimization engine avoids impractical or operationally unfeasible outcomes. Table 3 outlines a broad range of objectives that can be addressed using the optimization tool. The insights gained from these case studies have played a key role in refining the steam distribution strategy, promoting more efficient and data-driven decision-making within the South Ratqa project.

Table 3—Description of Optimization Case Studies and Their Objectives

Case	Description	Objective
1	CSS Patterns Selection, Excluding 11 New Patterns	A scenario in which the steam distribution process was optimized based on the exclusion of the 11 new patterns, focusing on the existing patterns' requirements.
2	CSS Patterns Selection, Including 11 New Patterns	An evaluation of how the inclusion of the 11 new patterns impacts the steam distribution process, ensuring that the additional patterns were effectively integrated into the overall steam plan.
3	Optimal Number of Sequential CSS Cycles, for the 11 New Patterns	This case study addressed the determination of the most efficient number of sequential CSS cycles required for the 11 new patterns, considering the constraints and objectives of steam utilization.
4	Maximum SF Conversions and Candidate Patterns, for SF Conversion	This case study aimed to determine the maximum number of patterns that could be converted from CSS to SF mode, identifying the candidate patterns suitable for such a conversion based on performance and capacity.
5	Optimal Steam Injection Rate, for SF Patterns	Focused on calculating the optimal steam injection rate for patterns operating in SF mode, ensuring that the injection rate was aligned with production targets and operational constraints.
6	Minimum Steam Requirement, for All Patterns	This case study evaluated the minimum steam required for all active patterns, aiming to ensure that each pattern received absolute minimum steam to meet production targets while minimizing overall steam consumption.

Further elaboration on using the tool is provided by expanding Case 4 in the following section.

Maximum SF Conversions and Candidate Patterns for SF Conversion

To mitigate the significant reservoir depletion observed in older CSS patterns, two patterns were converted from Cyclic Steam Stimulation (CSS) to Steam Flood (SF) in August 2022. The encouraging uplift in oil production following these initial conversions led to the decision to convert six additional patterns to SF in 2023. These positive results raised an important operational question: Is it more advantageous to convert additional patterns to SF—thereby limiting the number of patterns receiving steam stimulation—or to continue CSS operations across a broader set of patterns?

Assuming that each SF pattern requires approximately 55% more steam than a CSS pattern, the steam optimization tool indicated that converting more patterns to SF yields better production outcomes than maintaining CSS operations on the older patterns. Under a steam capacity of 100 MBCWEPD, the optimization tool identified that the most favorable strategy included: steam stimulation of the 11 new patterns, converting two additional patterns in Block 3 to SF, stimulating one older CSS pattern, and producing the remaining patterns under cold mode. This configuration allowed for more effective utilization of the available steam capacity, optimizing short-term production.

Furthermore, while the incremental oil gain from converting more patterns to SF was relatively modest, there is a significant operational and cost-saving advantages. SF eliminates the repeated well interventions required in CSS operations—such as the cyclical conversion of wells between injection and production—which reduces operational complexity and associated costs. These factors enhance the overall economic viability of SF, making it a more sustainable and efficient option in the long term.

Summary and Conclusions

This study presents an efficient and time-saving methodology for steam distribution across South Ratqa patterns and helps selecting the optimal combination of operating modes to maximize short-term production. The process begins by extracting historical injections and production data from the OFM database, followed by generating three distinct production forecasts for each pattern—Cold, CSS, and SF—based on historical performance and reservoir properties.

An optimization tool is then employed to identify the optimal combination of operating modes that maximizes production while respecting well and facility constraints and staying within the available steam capacity. This enables the development of a steam injection plan along with a corresponding short-term production forecast across multiple scenarios, significantly accelerating the steam optimization process.

The automated workflow effectively addresses the challenge of time-intensive simulation-based forecasting, allowing reservoir engineers to rapidly identify the most effective steam plan. By integrating forecasting and optimization into a single platform, the tool significantly reduces the time required for scenario evaluation and decision-making, supporting both short- and mid-term production planning.

The developed steam optimization tool was successfully applied to determine the optimal short-term steam allocation strategy under limited steam capacity conditions and to meet the steam requirements across all active patterns in the South Ratqa Field, Kuwait.

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